KS3 DIGITAL TECHNOLOGY

GENERATIVE AI

How AI tools create text, images, audio and video, how to write good prompts and check the results, and how to use generative AI honestly, safely and well.

LESSONS

1 What is generative AI?
2 How chatbots write
3 Prompting well
4 Hallucinations and bias
5 Using AI honestly
6 AI, society and the future

LESSON 1 OF 6

WHAT IS GENERATIVE AI?

Generative AI is artificial intelligence that creates new content (text, images, audio, video or code) after learning patterns from huge amounts of examples.

DO NOW

Name three things a computer can make for you today that it could not make ten years ago.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what generative AI is and give examples of what it can create

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what makes AI "generative"
- ✓ I can give examples of text, image, audio and video generators
- ✓ I can explain the difference between generative AI and other AI

AI THAT MAKES THINGS

- Most AI you meet sorts or predicts: is this email spam? which video next?
- Generative AI creates something new: a story, a picture, a song, some code
- It learns patterns from millions of examples, then makes new content that fits those patterns
- You ask it with a prompt: an instruction typed or spoken in everyday language

KEY WORDS

GENERATIVE AI	Artificial intelligence that creates new content such as text, images, audio, video or code.
TRAINING DATA	The huge collection of examples (text, images, sounds) that an AI model learns patterns from.
MODEL	The result of training an AI: a very large set of numbers that captures the patterns it has learned.
PROMPT	The instruction you give a generative AI tool, usually in everyday language.

ACTIVITY · 10 MINUTES · PAIRS

SORT OR CREATE?

1. Read the list of ten everyday AI uses your teacher shows on the board.
2. Decide for each one: does the AI sort/predict, or does it create something new?
3. Write your answers in the box, then compare with another pair.

In your workbook, answer:

? List the uses you decided were generative AI.

? Pick one example you disagreed about. Why was it tricky?

Your workbook: Lesson 1 · tick each step off as you go

WHAT CAN IT CREATE?

- Text: chatbots and writing assistants answer questions, summarise and draft
- Images: describe a picture and it draws it in any style
- Audio: speech in someone's voice, music in any genre
- Video: short clips from a description, or edits to real footage
- Code: whole programs from a description (lesson 2 of "Coding with an AI assistant")

HOW DID IT LEARN?

- Training data: huge collections of text, images and sounds, much of it from the internet
- The model is the result of training: a giant set of numbers that captures patterns
- It does not store a copy of every example: it learns what usually goes together
- More data and more computing power have made today's models far more capable

QUICK CHECK

1. What makes AI "generative"?

A It sorts things into groups
B It only works on robots
C It creates new content such as text or images
D It searches the internet for answers

2. Which of these is an example of generative AI?

A A calculator
B An app that draws a picture from a description
C Face unlock on a phone
D A spam filter

3. What is training data?

A The instructions you type into a chatbot
B The answer the AI gives you
C The computer the AI runs on
D The examples an AI learns patterns from

4. What do we call the instruction you give to a generative AI tool?

A A prompt
B A model
C A dataset
D An algorithm

QUICK CHECK: ANSWERS

1. C: It creates new content such as text or images

Generative AI makes new content rather than just labelling or choosing.

2. B: An app that draws a picture from a description

Making a new picture is generating content.

3. D: The examples an AI learns patterns from

Models learn from huge collections of examples called training data.

4. A: A prompt

A prompt is the instruction, usually in everyday language.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between generative AI and AI that sorts or predicts. Give an example of each.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what makes AI "generative"
- ✓ I can give examples of text, image, audio and video generators
- ✓ I can explain the difference between generative AI and other AI

LESSON 2 OF 6

HOW CHATBOTS WRITE

Chatbots are powered by large language models.

GENERATIVE AI

DO NOW

Finish the sentence: "Once upon a ..." How did you know what word comes next?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how a large language model writes by predicting the next word

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what a large language model (LLM) is
- ✓ I can describe how an LLM writes one token at a time
- ✓ I can explain why fluent writing is not the same as being right

PREDICT THE NEXT WORD

- "Once upon a ..." you already know the answer: time
- You learned that pattern from hearing thousands of stories
- A large language model (LLM) does the same with billions of sentences
- It writes by predicting the next small chunk of text, over and over

KEY WORDS

LARGE LANGUAGE MODEL	A generative AI model trained on huge amounts of text that writes by predicting the next token.
TOKEN	A small chunk of text (a word or part of a word) that a language model reads and writes.
CHATBOT	A program you can have a conversation with, often powered by a large language model.

TOKENS

- Text is broken into tokens: whole words or pieces of words
- "unbelievable" might be split into "un", "believ", "able"
- The model gives every possible next token a probability
- It picks one, adds it to the text, and repeats until the answer is finished

ACTIVITY · 15 MINUTES · GROUPS OF 3

BE THE LANGUAGE MODEL

1. Your teacher gives each group a sentence starter and a bag of word cards.
2. Take turns to choose the most likely next word card and add it to the sentence.
3. After five words, choose one card that is unlikely but still makes sense. How does the sentence change?
4. Write your two sentences below.

In your workbook, answer:

? Write your "most likely" sentence and your "surprising" sentence.

? Explain how this game is like the way an LLM writes.

Your workbook: Lesson 2 · tick each step off as you go

FLUENT IS NOT THE SAME AS TRUE

- The model is choosing likely words, not checking facts
- So it can sound confident and still be wrong
- It can also be "creative" on purpose: a setting called temperature adds variety
- Good users check important facts somewhere reliable

QUICK CHECK

1. What does LLM stand for?

A Long logic machine
B Linked list model
C Learning language memory
D Large language model

2. How does an LLM write an answer?

A By asking a human to type it
B By looking it up in a dictionary
C By predicting the next token again and again
D By copying a full answer from a website

3. What is a token?

A A coin used to pay for AI
B A picture made by AI
C A password for the chatbot
D A word or part of a word that the model works with

4. Why can a chatbot give a wrong answer that sounds right?

A It picks likely words rather than checking facts
B It is designed to lie
C It only reads old newspapers
D It cannot spell

QUICK CHECK: ANSWERS

1. D: Large language model

An LLM is a large language model.

2. C: By predicting the next token again and again

It builds the answer one predicted token at a time.

3. D: A word or part of a word that the model works with

Text is split into tokens: words or pieces of words.

4. A: It picks likely words rather than checking facts

Fluent text comes from patterns, not fact-checking.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe how a large language model produces an answer to a question.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what a large language model (LLM) is
- ✓ I can describe how an LLM writes one token at a time
- ✓ I can explain why fluent writing is not the same as being right

LESSON 3 OF 6

PROMPTING WELL

The quality of what generative AI gives you depends on the prompt.

STARTER · 5 MINUTES

DO NOW

Ask a friend to draw "an animal". Then ask for "a grumpy orange cat wearing a tiny hat, cartoon style". Which drawing will be closer to what you wanted?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Write clear prompts and improve them step by step to get better results

SUCCESS CRITERIA: I CAN...

- ✓ I can explain why vague prompts give poor results
- ✓ I can use a prompt recipe: role, task, context, format, limits
- ✓ I can improve a prompt by refining it

VAGUE IN, VAGUE OUT

- "Tell me about volcanoes" could mean anything
- The AI has to guess your age, purpose and how much you want
- Better prompts remove the guessing
- Writing good prompts is called prompt engineering

KEY WORDS

PROMPT ENGINEERING	The skill of writing prompts that get useful results from a generative AI tool.
CONTEXT	Background information that helps an AI (or a person) understand what you need.
ITERATION	Repeating a cycle of trying, checking and improving.

A PROMPT RECIPE

- Role: "You are a geography teacher for 12-year-olds"
- Task: "Explain how volcanoes form"
- Context: "for a revision page; we've learned about tectonic plates"
- Format: "in 5 bullet points with one key word in bold each"
- Limits: "under 120 words; use simple language"

KEEP IMPROVING

- Read the answer critically: is it what you need?
- Refine: add detail, ask for a change, give an example
- Follow up: "Make point 3 simpler" or "Give me a real example"
- This loop of trying and improving is called iteration

ACTIVITY · 20 MINUTES · INDIVIDUAL

PROMPT MAKEOVER

1. Use your school-approved AI tool (or the printed answers your teacher shows if you can't use one).
2. Type the vague prompt: "Write about the water cycle."
3. Rewrite it using the recipe: role, task, context, format and limits. Try it.
4. Refine it once more with a follow-up instruction.

In your workbook, answer:

? Write your improved prompt.

? How was the second answer better than the first?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. Why does a vague prompt often give a poor answer?

A The AI has to guess what you want
B The AI refuses vague prompts
C The AI only answers questions in capitals
D Vague prompts use more electricity

2. Which part of a prompt says how the answer should be laid out?

A Role
B Limits
C Context
D Format

3. What is iteration when prompting?

A Turning the AI off and on
B Asking the same thing in capitals
C Trying, checking and improving the prompt
D Copying someone else's prompt

4. Which of these should you never put into a prompt?

A The age of your audience
B Your home address
C The topic you are studying
D The word limit you need

QUICK CHECK: ANSWERS

1. **A: The AI has to guess what you want**
Without detail, the AI guesses your age, purpose and format.

2. **D: Format**
Format is about layout, e.g. bullet points or a table.

3. **C: Trying, checking and improving the prompt**
Iteration means improving the prompt in cycles.

4. **B: Your home address**
Personal information can be stored by the AI tool.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Rewrite this prompt to make it better, and explain one improvement: "Make a quiz."

Look back at today's success criteria: can you tick them all?

- ✓ I can explain why vague prompts give poor results
- ✓ I can use a prompt recipe: role, task, context, format, limits
- ✓ I can improve a prompt by refining it

LESSON 4 OF 6

HALLUCINATIONS AND BIAS

Generative AI can "hallucinate", inventing facts, quotes or sources, and it can repeat biases found in its training data.

STARTER · 5 MINUTES

DO NOW

A chatbot says "Enniskillen Castle was built in 1066 by William the Conqueror." How could you check whether that's true?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why generative AI makes mistakes and shows bias, and check its output

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what an AI hallucination is
- ✓ I can explain how bias gets into an AI model
- ✓ I can check AI output using reliable sources

WHEN AI MAKES THINGS UP

- A hallucination is when AI states something false as if it were true
- It can invent facts, dates, quotes, people and even sources or web links
- It happens because the model predicts likely words, not true ones
- Hallucinations look exactly as confident as correct answers

KEY WORDS

HALLUCINATION	When a generative AI states something false as if it were true, such as an invented fact or source.
BIAS	When a system treats some people or ideas unfairly or one-sidedly, often because of patterns in its data.
VERIFY	To check that something is true, for example using reliable sources.

BIAS IN, BIAS OUT

- Training data comes from people, so it carries people's biases
- If most pictures of "a scientist" in the data are men, the AI may draw mostly men
- Some languages, cultures and places are under-represented
- Bias can be unfair, hurtful or simply wrong

CHECK BEFORE YOU TRUST

- Is it important? The more it matters, the more you check
- Cross-check facts with two reliable sources (BBC Bitesize, gov.uk, textbooks)
- Click any links: do they exist and say what the AI claims?
- Ask: whose view is missing?

ACTIVITY · 20 MINUTES · PAIRS

FACT-CHECK THE CHATBOT

1. Your teacher shows five statements written by an AI chatbot about Northern Ireland.
2. For each one, use two reliable websites to decide: true, false or partly true.
3. Record the sources you used.

In your workbook, answer:

? Which statements were hallucinations, and what was actually true?

? Which sources did you trust, and why?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is an AI hallucination?

A When AI crashes
B When AI shows a picture
C When AI states something false as if it were true
D When AI asks you a question

2. Where does bias in AI usually come from?

A The keyboard you use
B The colour of the screen
C The speed of the internet
D Patterns in the training data

3. What is the best way to verify a fact from a chatbot?

A Ask a friend who agrees with you
B Ask the chatbot if it is sure
C Trust it if it sounds confident
D Check it with two reliable sources

4. An AI gives you a link to a source. What should you do?

A Trust it because it has a link
B Open it and check it says what the AI claims
C Ignore all links
D Copy it into your homework straight away

QUICK CHECK: ANSWERS

1. C: When AI states something false as if it were true

Hallucinations are confident but false outputs.

2. D: Patterns in the training data

AI learns biases that exist in its training data.

3. D: Check it with two reliable sources

Cross-checking with reliable sources is the best test.

4. B: Open it and check it says what the AI claims

AI can invent links and sources, so open them.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain what is meant by an AI hallucination.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what an AI hallucination is
- ✓ I can explain how bias gets into an AI model
- ✓ I can check AI output using reliable sources

LESSON 5 OF 6

USING AI HONESTLY

Using AI can be helpful or dishonest depending on how you use it and whether you say so.

STARTER · 5 MINUTES

DO NOW

Is it cheating to use a calculator in a maths test? Is it cheating to use AI to write your English homework? What's the difference?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Use generative AI honestly in schoolwork and respect copyright and ownership

SUCCESS CRITERIA: I CAN...

- ✓ I can tell the difference between using AI to learn and using it to cheat
- ✓ I can acknowledge when I have used AI
- ✓ I can explain copyright questions raised by AI-generated content

HELPER OR GHOST-WRITER?

- Using AI to explain, quiz you or give feedback can help you learn
- Getting AI to do the work and handing it in as yours is plagiarism
- Exams and coursework have strict rules: breaking them can mean losing marks or the whole qualification
- When in doubt, ask your teacher and always say what you used

KEY WORDS

PLAGIARISM	Presenting someone else's work (including work made by AI) as your own.
COPYRIGHT	The legal right that protects creative work such as writing, music, photos and art from being copied without permission.
ACKNOWLEDGE	To say clearly where something came from, including which AI tool you used and how.

SAYING WHAT YOU USED

- Acknowledge AI like any other source
- Name the tool, the date, what you asked and how you used the answer
- Example: "I used Copilot on 3 March to suggest ideas for my plan, then wrote it myself"
- Your school's AI rules come first

ACTIVITY · 15 MINUTES · GROUPS OF 4

HONEST OR NOT?

1. Your group gets six scenario cards about pupils using AI.
2. Place each card on a line from "completely honest" to "cheating".
3. Agree a rule your class could use for using AI in homework.

In your workbook, answer:

? Write your group's rule for using AI in homework.

? Choose the hardest scenario. Explain where you placed it and why.

Your workbook: Lesson 5 · tick each step off as you go

WHO OWNS AI-MADE WORK?

- AI is trained on work made by real artists, writers and musicians
- Many creators say their work was used without permission or payment
- Laws about AI and copyright are still being argued about in courts and governments
- Never use AI to copy a real artist's style or voice and pretend it's theirs

QUICK CHECK

1. What is plagiarism?

A Reading a library book
B Using a spellchecker
C Presenting someone else's work as your own
D Asking a teacher for help

2. Which use of AI is most likely to help you learn honestly?

A Using it secretly in a test
B Asking it to quiz you on a topic
C Copying its answer into coursework
D Asking it to write your essay and handing it in

3. What should you include when you acknowledge using AI?

A The tool, date, what you asked and how you used it
B The AI's password
C Nothing, if the answer was good
D Only your name

4. Why is copyright an issue for generative AI?

A AI cannot make pictures
B Models were trained on people's creative work, often without permission
C AI work is always free to use
D Copyright only applies to robots

QUICK CHECK: ANSWERS

1. C: Presenting someone else's work as your own

Plagiarism is passing off others' work as yours, including AI work.

2. B: Asking it to quiz you on a topic

Quizzing yourself with AI supports your own learning.

3. A: The tool, date, what you asked and how you used it

Acknowledging means saying clearly how AI was used.

4. B: Models were trained on people's creative work, often without permission

Creators argue their work was used without consent or payment.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Give one way a pupil could use generative AI to help them learn, and one way that would be dishonest.

Look back at today's success criteria: can you tick them all?

- ✓ I can tell the difference between using AI to learn and using it to cheat
- ✓ I can acknowledge when I have used AI
- ✓ I can explain copyright questions raised by AI-generated content

LESSON 6 OF 6

AI, SOCIETY AND THE FUTURE

Generative AI is changing work, learning and creativity.

STARTER · 5 MINUTES

DO NOW

Name one job you think AI will change a lot in the next ten years. Will it make the job better or worse?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Discuss the impact of generative AI on jobs, the environment and society, and form a reasoned opinion

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how generative AI may change jobs
- ✓ I can explain the environmental cost of AI
- ✓ I can give a reasoned opinion about a future AI issue

JOBS: CHANGE, NOT JUST LOSS

- Some tasks will be automated: drafting, summarising, basic coding, translation
- New jobs appear: AI trainers, AI safety testers, prompt designers, AI auditors
- Many jobs change rather than vanish: people work with AI tools
- Human skills matter more: creativity, judgement, care, teamwork

KEY WORDS

AUTOMATION	When machines or software do tasks that people used to do.
REGULATION	Official rules, for example laws that control how AI may be used.
DATA CENTRE	A building full of computers (servers) that store data and run online services such as AI.

THE HIDDEN COST

- AI models are trained and run in huge data centres full of computers
- These use a lot of electricity and water for cooling
- Training one large model can use as much electricity as many homes use in a year
- Choosing when AI is really needed is part of using it responsibly

WHO DECIDES?

- Governments are creating rules (regulation), e.g. the EU AI Act
- Companies publish their own safety rules and test models before release
- Schools and exam boards set rules for AI in learning
- Your voice matters: you'll grow up with these tools

ACTIVITY · 25 MINUTES · WHOLE CLASS

CLASS DEBATE: "AI WILL DO MORE GOOD THAN HARM"

1. Your teacher splits the class into "for" and "against".
2. In your group, find three strong points with an example for each.
3. Debate! Each side gets two minutes, then one minute to respond.
4. Afterwards, write your own opinion, even if it differs from your side.

In your workbook, answer:

? Write your best argument for your side, with an example.

? Now write your own opinion in 3-4 sentences, giving reasons.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is automation?

A Machines or software doing tasks people used to do
B Writing a prompt
C Turning a computer off
D Making a car go faster

2. Why does generative AI have an environmental cost?

A Data centres use a lot of electricity and water
B AI uses paper to print answers
C It doesn't have one
D It needs petrol to run

3. What is regulation?

A A data centre
B Official rules set to control something
C A type of computer
D A kind of prompt

4. Which skill is likely to become MORE important as AI improves?

A Copying text quickly
B Typing very fast
C Judging whether information is right
D Memorising phone numbers

QUICK CHECK: ANSWERS

1. A: Machines or software doing tasks people used to do

Automation means tasks done by machines instead of people.

2. A: Data centres use a lot of electricity and water

Running and training models takes energy and cooling.

3. B: Official rules set to control something

Governments create regulation to control how AI is used.

4. C: Judging whether information is right

Checking and judging AI output is a key human skill.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain one way generative AI could change jobs.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how generative AI may change jobs
- ✓ I can explain the environmental cost of AI
- ✓ I can give a reasoned opinion about a future AI issue

END OF TOPIC

GENERATIVE AI

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.